OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

# EXP - 5: SCHEDULING ALGORITHMS (Round Robin Scheduling)

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```
user123@localhost:~/scheduling
[user123@localhost ~]$ ls
abc.sh  awk  Desktop  Downloads  ifelse.sh  memory  page  Pictures  Public  REC  sample.sh  scheduling  SFA  system_calls  Videos
ab.sh  bankers  Documents  Folder  loops.sh  Music  pc  posneg.sh  qns.txt  sample.c  sample.txt  script  sh  Templates  while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
```

```
user123@localhost:~/scheduling — /usr/bin/vim rr.c
#include <stdio.h>
int main(){
        int n, tq, completed = 0, completionTime = 0;

        printf("Enter number of processes: \n");
        scanf("%d", &n);

        printf("Enter Time Quantum: \n");
        scanf("%d", &tq);

        int at[n], bt[n], rem[n], tat[n], wt[n];
        float avgtat = 0, avgwt = 0;
        // Input
        for (int i = 0; i < n; i++)
        {
                printf("Enter AT and BT for P%d: \n", i + 1);
                scanf("%d %d", &at[i], &bt[i]);
                rem[i] = bt[i]; // Remaining time = Burst time
        }
        // Round Robin Logic
        while (completed < n)
        {
                int idle = 1;
                for (int i = 0; i < n; i++)
                {
                        if (at[i] <= completionTime && rem[i] > 0) //
                        {
                                idle = 0; //
                                if (rem[i] > tq) //3,5 /1>2
                                {
                                        completionTime += tq; //1+2 ==3
                                        rem[i] -= tq; //5-2=3
                                } else {
                                        completionTime += rem[i]; //
                                        tat[i] = completionTime - at[i]; // Turnaround Time
                                        wt[i] = tat[i] - bt[i]; // Waiting Time
                                        rem[i] = 0;
                                        completed++;
                                }
                        }

                }
                if (idle){
                        completionTime++; // If no process arrived yet//1
                }
        }

        // Calculate averages
        for (int i = 0; i < n; i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);

        printf("Average Waiting Time: %.2f\n", avgwt / n);

        return 0;
}
```

```
user123@localhost:~/scheduling

[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Vi
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     wh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
[user123@localhost scheduling]$ gcc rr.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 4
Enter AT and BT for P3:
2 2
Enter AT and BT for P4:
3 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
```